

Assignments 5 (Chapter 6) Solutions

1. (Trappe, page 159: 1) The ciphertext 5859 was obtained from the RSA algorithm using $n = 11413$ and $e = 7467$. Using the factorization $11413 = 101 \cdot 113$, find the plaintext

Sol:

$$\phi(n) = (p-1) \cdot (q-1) = 100 \cdot 112 = 11200$$

$$d \equiv e^{-1} \equiv 3 \pmod{11200} \text{ (using extended Euclidean algorithm)}$$

$$m \equiv c^d \equiv 5859^3 \equiv 1415 \pmod{11413}$$

$$\text{verification: } c \equiv 1415^{7467} \equiv 5859 \pmod{11413}$$

2. (Trappe, page 159: 4) Naïve Nelson uses RSA to receive a single ciphertext c , corresponding to the message m . His public modulus is n and his public encryption exponent is e . Since he feels guilty that his system was used only once, he agrees to decrypt any ciphertext that someone sends him, as long as it is not c , and return the answer to that person. Evil Eve sends him the ciphertext $2^e c \pmod{n}$. Show how this allows Eve to find m .

Sol:

Nelson checks that $2^e c$ is not c , then decrypts it and gets

$$(2^e c)^d \equiv 2^e c^d \equiv 2^e m \pmod{n}$$

Eve divides this by 2 and obtains the original message m .

3. (Trappe, page 159: 5) In order to increase security, Bob chooses n and two encryption exponents e_1, e_2 . He asks Alice to encrypt her message m to him by first computing $c_1 \equiv m^{e_1} \pmod{n}$, then encrypting c_1 to get $c_2 \equiv c_1^{e_2} \pmod{n}$. Alice then sends c_2 to Bob. Does this double encryption increase security over single encryption? Why or why not?

Sol:

Alice encrypts her message m as $c_2 \equiv c_1^{e_2} \equiv m^{e_1 e_2} \pmod{n}$

To decrypt this message, Bob calculates $(c_2^{d_2})^{d_1} \pmod{n}$, where $e_1 \cdot d_1 \equiv 1 \pmod{\phi(n)}$ and $e_2 \cdot d_2 \equiv 1 \pmod{\phi(n)}$.

However, the above procedure is equivalent to the following

Alice encrypts her message m as $c_2 \equiv m^{e_3} \pmod{n}$ and

Bob decrypts the ciphertext as $m \equiv c_2^{d_3} \pmod{n}$ where

$$e_1 \cdot e_2 \equiv e_3 \pmod{\phi(n)} \text{ and } d_1 \cdot d_2 \equiv d_3 \pmod{\phi(n)}$$

(which naturally leads to $e_3 \cdot d_3 \equiv 1 \pmod{\phi(n)}$ and

$$c_2^{d_3} \equiv (m^{e_3})^{d_3} \equiv m \pmod{n})$$

Therefore, this scheme is equivalent to a simple RSA scheme with a different key pairs $((n, e_3), (n, d_3))$. The security is not enhanced.

4. (Trappe, page 159: 6) Let p and q be distinct odd primes, and let $n = p \cdot q$. Suppose that the integer x satisfies $\gcd(x, p \cdot q) = 1$.

(a) Show that $x^{\phi(n)/2} \equiv 1 \pmod{p}$ and $x^{\phi(n)/2} \equiv 1 \pmod{q}$

(b) Use (a) to show that $x^{\phi(n)/2} \equiv 1 \pmod{n}$

(c) Use (b) to show that if $e \cdot d \equiv 1 \pmod{\phi(n)/2}$ then $x^{e \cdot d} \equiv x \pmod{n}$ (This shows that we could work with $\phi(n)/2$ instead of $\phi(n)$ in RSA.)

Sol:

(a)

$$\phi(n)/2 = (p-1) \cdot (q-1)/2$$

$$2 \mid (q-1) \Rightarrow (p-1) \mid \phi(n)/2 \Rightarrow \phi(n)/2 = k \cdot (p-1)$$

therefore, $x^{\phi(n)/2} \equiv (x^k)^{p-1} \equiv 1 \pmod{p}$ from Fermat theorem

this is also true for $x^{\phi(n)/2} \equiv 1 \pmod{q}$

(b) using CRT,

$$x^{\phi(n)/2} \equiv 1 \pmod{p} \text{ and } x^{\phi(n)/2} \equiv 1 \pmod{q} \text{ implies } x^{\phi(n)/2} \equiv 1 \pmod{n}$$

Actually, this can also be derived from Carmichael theorem

$$x^{\lambda(n)} \equiv 1 \pmod{n} \text{ where } \lambda(n) = \text{lcm}(p-1, q-1)$$

since

$$\phi(n) = (p-1) \cdot (q-1) = \gcd(p-1, q-1) \cdot \text{lcm}(p-1, q-1) \text{ and}$$

$$2 \mid \gcd(p-1, q-1)$$

$$\text{i.e. } \text{lcm}(p-1, q-1) \mid \phi(n)/2$$

(c) if $e \cdot d \equiv 1 \pmod{\phi(n)/2}$ then

$$e \cdot d = 1 + k \cdot \phi(n)/2$$

$$x^{e \cdot d} \equiv x^{1 + k \cdot \phi(n)/2} \equiv x \pmod{n}$$

from the result of (b)

5. (Trappe, page 160: 12) Show that if $x^2 \equiv y^2 \pmod{n}$ and $x \not\equiv \pm y \pmod{n}$, then $\gcd(x+y, n)$ is a nontrivial factor of n

Sol:

We prove this in a way exactly the same as we prove the basic principle of factoring.

let $d = \gcd(x+y, n)$

There are three cases to be discussed:

Case 1 $d = n$: if $d = n$, then $x+y \equiv 0 \pmod{n}$, contradiction.

Case 2 $d = 1$:

$$\begin{aligned}x^2 &\equiv y^2 \pmod{n} \Rightarrow x^2 - y^2 = (x-y)(x+y) = k \cdot n \\&\Rightarrow x-y \equiv 0 \pmod{n}, \text{ contradiction}\end{aligned}$$

6. (Trappe, page 162: 4) Factor 618240007109027021 by the p-1 method

Sol:

Because this number is bigger than 2^{32} , MATLAB cannot process this number directly. We are going to use MAPLE kernel in MATLAB for solving this factoring problem with p-1 factoring method.

In a p-1 factoring method, we hope that there is discrepancy in the factors of p-1 and q-1.

If we can find a number, say $B!$, such that $p-1 \mid B!$ but $q-1 \nmid B!$, then $a^{B!} \equiv 1 \pmod{p}$ but $a^{B!} \not\equiv 1 \pmod{q}$, i.e. $a^{B!} - 1 = k \cdot p \neq k' \cdot q$. Therefore, $a^{B!} - 1 \pmod{n}$ is not 0 and $\gcd(a^{B!} - 1 \pmod{n}, n) = p$ is a nontrivial factor of n .

```
>> maple('n:=618240007109027021; B:=5; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 5   p := 1
>> maple('n:=618240007109027021; B:=10; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 10   p := 1
>> maple('n:=618240007109027021; B:=15; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 15   p := 1
>> maple('n:=618240007109027021; B:=20; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 20   p := 1
>> maple('n:=618240007109027021; B:=25; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 25   p := 250387201
>> maple('n:=618240007109027021; B:=30; p:=gcd((5&^factorial(B)) mod n -1, n)')
ans =      n := 618240007109027021   B := 30   p := 250387201
```

As a comparison, we factors 618240007109027021 directly with ifactor():

```
> maple('ifactor(618240007109027021)')
ans = ``(2469135821)*``(250387201)
```

If you have the MAPLE software installed, you can write a for loop to perform the above search procedure, the codes looks like:

```
n := 618240007109027021;
B := 1;
```

```

a := 5;
for i from 1 to 30 do
    B = B * i;
    p:=gcd((a&^B) mod n -1, n)
od;

```

7. (Trappe, page 162: 6) Let $n = 537069139875071$. Suppose you know that $85975324443166^2 \equiv 462436106261^2 \pmod{n}$

Factor n .

Sol:

Clearly $85975324443166 \equiv \pm 462436106261 \pmod{n}$ because

$$-462436106261 \equiv 536606703768810 \pmod{n}$$

$$\gcd(85975324443166 - 462436106261, 537069139875071) = 9876469$$

$$\gcd(85975324443166 + 462436106261, 537069139875071) = 54378659$$

$$\text{while } 9876469 * 54378659 = 537069139875071$$

the above can be easily done through MAPLE

```

>> maple('n:= 537069139875071; b:=n - 462436106261;')
ans =  n := 537069139875071  b := 536606703768810
>> maple('c=gcd(537069139875071, 85975324443166-462436106261)')
ans =  c = 9876469
>> maple('c=gcd(537069139875071, 85975324443166+462436106261)')
ans =  c = 54378659
>> maple('c=9876469*54378659')
ans =  c = 537069139875071

```